

Core Electronics Projects

FPGA-Based UART Transceiver Design | Personal Project [Month'YY - Month'YY]

- Designed a UART transmitter and receiver module in Verilog HDL supporting configurable baud rates for asynchronous serial communication [\[Code / Proof\]](#)
- Simulated the design using a self-checking testbench in ModelSim/Vivado, verifying correct framing, start/stop bits, and parity handling [\[Code / Proof\]](#)
- Synthesized and implemented the design on an FPGA (Basys3/Artix-7), validating real-time transmission with a UART-to-PC terminal interface [\[Code / Proof\]](#)
- Optimized the design for minimal resource utilization and analyzed timing constraints to ensure reliable operation at higher baud rates [\[Code / Proof\]](#)

Note: GitHub links are placeholders — replace 'yourusername/repo-name' with your actual repository paths, and fill in the project dates once finalized.